15th Oct 2022

(1) RIDGE

(2) LASSO

(3) ELASTIC REGRESSION

(4) LINEAR-REGRESSION

Linear Regression

This below mention alogrithms or topic is Extension of 'Linear Regression' only. that we will continue.....

(1) Ridge Regression

(2) LASSO Regression

(3) Elastic Regression or Elasticnet Regression

(4) Linear Regression practical.

(1) Ridge Regression or (L2 Regularization) or (L2 Norm)

Ridge Regression also called as L2 Regularization. or (L2 Norm)

The main aim of Ridge Regression is basically to reduce overfitting.

* How ~~d~~ this reduce overfitting?

Let's try to understand it.

Let's say, I have these 2 data points in my training dataset.

If I try to create a 'best fit line.' It looks something like this, It passes the both the lines.

If it is passing through both the ~~...~~ points, Can we say this data is overfitted.

why we do say overfitted?

Can I say here my 'cost function' will be exactly zero ('0')

* with respect to ~~thing~~ this, what will be my ~~.~~ training data ~~.~~ interms of ~~.~~ 'BIAS'.

Overfitting

Training Data → [redacted] 'Low BIAS'

Can we say, this will be [redacted] 'Low' BIAS.

what is our overfitting condition, basically means our training data will have [redacted] 'Low BIAS'.

* with respect to 'Test data' what this [redacted] performance will be?

Let's say, we bring out our 'Test data' somewhere here & there. If I am having this kind of 'Test data' obviously the performance will be bad. If the performance bad, what we will say over here, interms of variance.

Test data → : Low or
High value

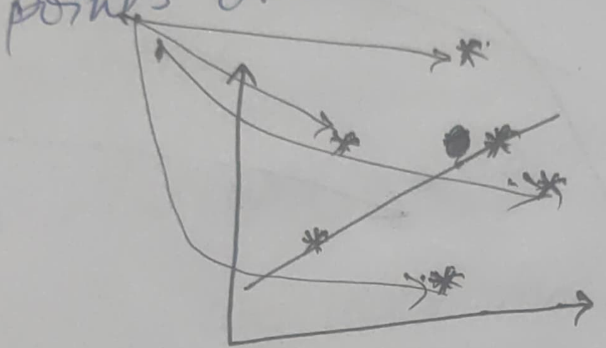
It can be, Low or High variance.
have both

* Training data : 'Low BIAS' because for the training data, the model is performing exceptionally well.

* Test data :- For test data, line may performed well, or it may not perform well.
Both the senario can happend.

~~Explaing~~.
Explaning again,

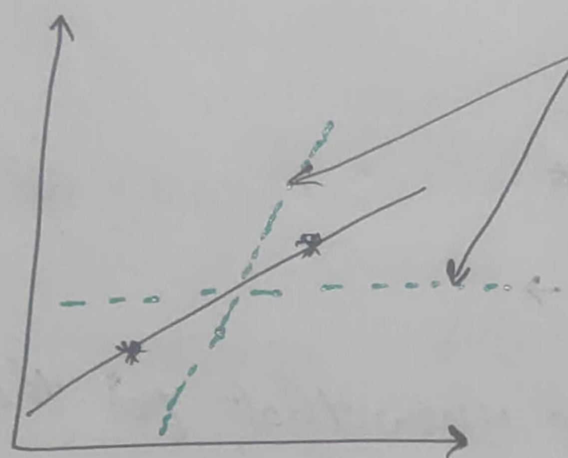
Let's say,
This is my 2 data point, we know that we will overfit this lines.
But what happend we get new data points over here.



In this senario, we know that, for the 'Training data' perfectly fitted.
For the 'Test data' it is not performing well. That is the region we have written as low or high variance.
That is the condition, we have given as 'overfitting'.

Now,

~~I~~ what if I try to do something else.
I try to make sure try to create a best fit line which looks like this, or like this



I try to create my best fit line, like this.
So we this Error ~~get~~ get reduce with respect to the 'Test data' also now.

Yes, right.

* But, In Linear Regression, we have understood that, we always going to get 'best fit line'.

But, To create this kind of line, we will be using the ~~[illegible]~~ something called as 'Ridge regression'.

'Ridge Regression' is used to reduce overfitting.

If we have this kind of senario we will try to reduce the overfitting condition.

* How will we do this?

we used the ~~san~~ same "cost function"

Cost function formula:

$$\frac{1}{m}\sum_{i=1}^{m}\left(h_\theta(x)^{(i)} - y^{(i)}\right)^2 + \lambda\sum_{i=1}^{m}(slope)^2$$

This is a my 'cost function' in 'linear Regression', we discussed this.

Along with this 'cost function', I will add 2 parameters (λ) lambda & another is $(slope)^2$.

How it is going to make sure that data points will be never overfitting.

This lambda (λ) is Hyper parameter & this (slope)^2 is nothing but slope of this line.

If Suppose, this is my equation

$$\boxed{h_\theta(x) = \theta_0 + \theta_1 x_1}$$

Then, my slope is nothing but my $(\theta_1)^2$

* Let's say,

my equation is like this

$$\boxed{h_\theta(x) = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_3 + \ldots \theta_n x_n}$$

Let's say,

I have 3 slopes, then what will be this (slope)^2 become.

Can I write $(\theta_1 + \theta_2 + \theta_3)^2$.

This the parameter, we have added.

* $\boxed{\lambda \cdot \sum_{i=1}^{m} (\text{slope})^2}$

what actually is this? let's understand.

If I give my Lambda value $\boxed{\lambda = 0}$
$\Downarrow$
$\boxed{\text{Linear Regression}}$

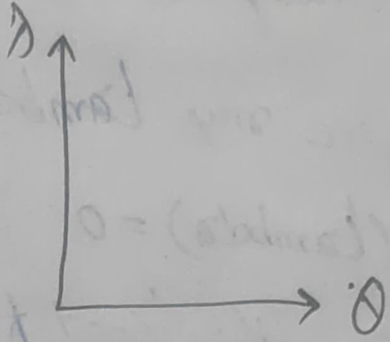
λ (Lambda) $= 0$

Can we say this 'cost function' will be the same like 'simple linear Regression' or Can we say my 'cost function' will be of a 'linear Regression' itself.

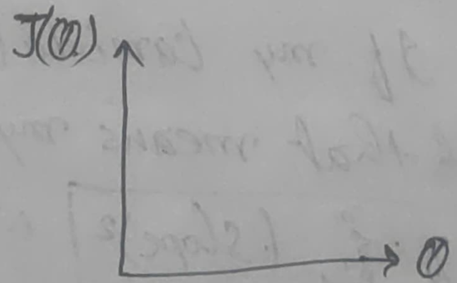
If my Lambda (λ) value is '0' zero, that means my this entire value $\rightarrow \boxed{\lambda \cdot \sum_{i=1}^{m} (\text{slope})^2}$ will become zero.

* what we will get? we get the 'cost function' of a 'linear Regression'.

* Question :- what is the ~~difference~~ relationship between the your slope & your Lambda (λ). ?



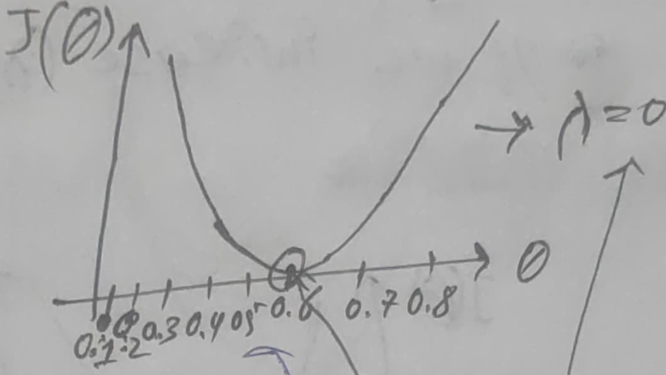
But,
1st I will write $J\theta$ or J of theta
$J\theta$ is a 'Cost function'.



(θ) is our Slope & $J(\theta)$ is our slope.

Now,
In this senario, my (λ) Lambda value is (0) zero.

what kind of curve, we will get.
Can, we write this kind of curve



let's say, this is my λ = -0.1, 0, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6... values.

* we are trying to find out that, what is relationship between this slope & Lambda (λ) value.

* Now, when my $\boxed{(\lambda) \text{ Lambda} = 0}$; do we will get 'Gradient desent curve' with this 'Global minima.'?

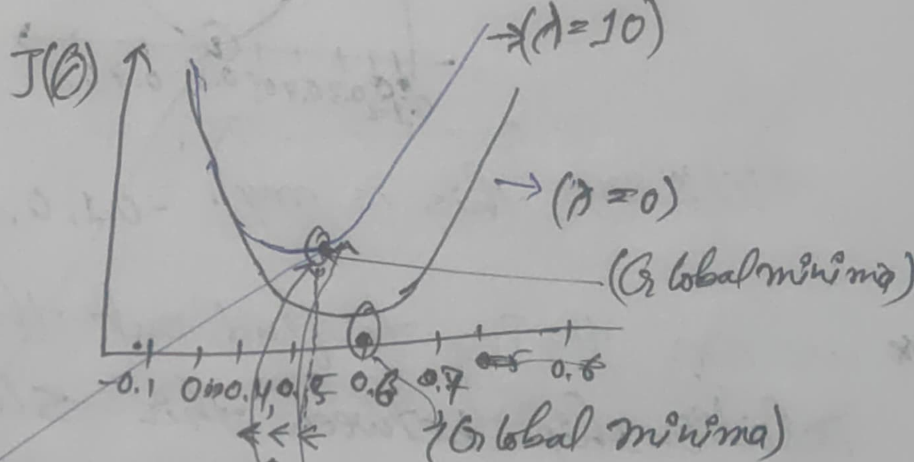
So, here can we write <u>(λ) Lambda = 0</u> right

Let's go ahead & increase the

$\boxed{\text{Lambda } (\lambda) = 10}$

If we increase $(\lambda) = 10$ then,

~~we can see that we~~

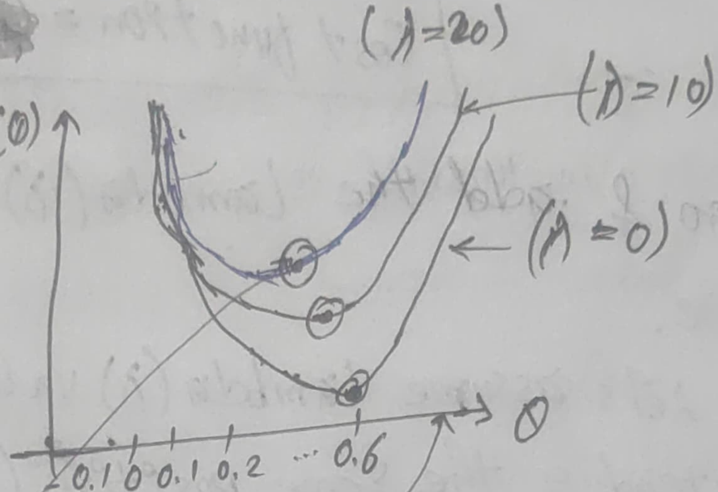


we can see that we will get the curve like this, If I get the curve like this what changes happened to the previous curve.

Can we say the change is that the 'Global minima' has shifted & the slope value also decrease.

This is is my Global minima; with respect to this the 'slope' has also decrease. This is my Co-effient over here.

If, again,
I will increase my $\boxed{\text{Lambda}(\lambda) = 20}$ again.



Then, here we will be see, it will go like this. $\boxed{\lambda = 20}$

In short, when I am decreasing my slope value, it is minimizing the 'Cost function'.

* For Example:-
when my $\boxed{\text{Lambda}(\lambda) = 0}$, I have my 'Cost Function' like this.
Let's consider same equation, I have this 2 data points & I have created my 'best fit line'.

then, what will be my cost function?

It will be zero(0) ~~[illegible]~~ initially,

$$\boxed{\text{Cost function} = 0}$$

* let's go & add the Lambda(λ) + slope over here.

let's assume lambda(λ) value is not $\boxed{10}$, instead of the some positive (+ve) value.

* Lambda(λ) = (+ve)
* $(\text{Slope})^2$ = (+ve)

Slope' since it is squaring up $\boxed{\underset{(+ve)}{\lambda} \sum_{i=1}^{m} \underset{(+ve)}{(\text{Slope})^2}}$

both this will be positive (+ve) value.

Note:- Lambda(λ) will never be negative, always positive (+ve) value.

So, Lambda(λ) $\times$ $(\text{Slope})^2$ will be always positive.

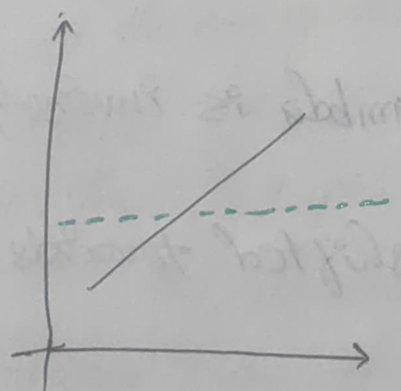
Cost function $= 0 + [+ve]$

$= +ve$

when, I increase this (+ve) value, this 'cost function' will be a positive (+ve) value.

If this is a positive value, do you feel that my line will be same place.

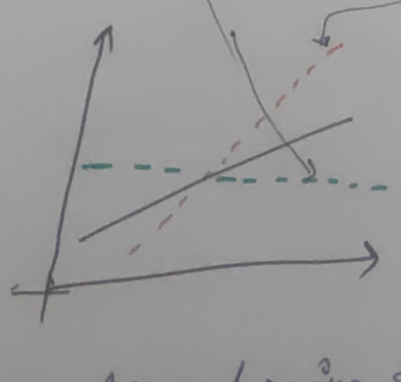
No, right my line will be switch.



Because, our main is to reduce the 'cost function'.

It can go up & (+ve)↓↓↓

It can go down, it can go up.



Since, I am having some positive (+ve) cost function value my linear regression knows that, I have to reduce it.

It will go ahead & change the <u>θ values.</u>

& It will go & try to create another 'best fit line.'

In this way, it will never get over fitted.

* <u>Note:</u>

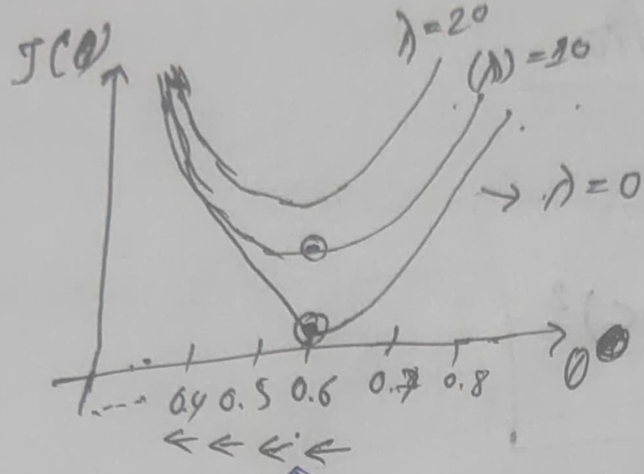
If I really want to talk about '(α) lambda' & 'slope'. Can I write (α) lambda is inversely proportional to the slope.

It will get shifted towards the left hand side.

<u>Question:</u>

why (α) lambda is inversely proportional to the slope. ?

Because when the Lambda (λ) value is increasing



Is this value get decreasing.

* So, at the end of the with Linear Regression we know that, we probably get a over fitted line. their is a possibility.

But, when we used "Ridge Regession", we always know that it never over fitted.

Question :

~~Eq~~ Equation

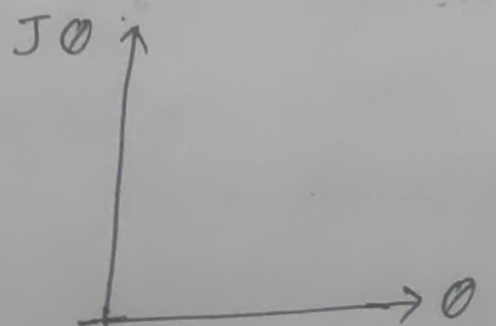
$$\boxed{\lambda \cdot \sum_{i=1}^{n} (\text{slope})^2}$$

hyper parameters

* what is the difference between Linear Regression & Ridge Regression ?
why we are using it.

we are using it to reduce the over fitting.

Let's draw a (Slope) θ & my Jθ ~~or~~ J(θ)



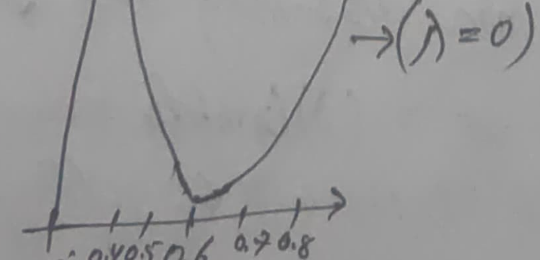
it include $\theta_1, \theta_2, \theta_3$ ....

Slope is nothing but (θ) thita & this is nothing but Jθ this is my 'Cost function'

Let's say,
If my (λ) Lambda is zero, which mean its a Linear Regression.
Because this entire value will become zero.

$$\boxed{\lambda \sum_{i=1}^{n} (slope)^2}$$

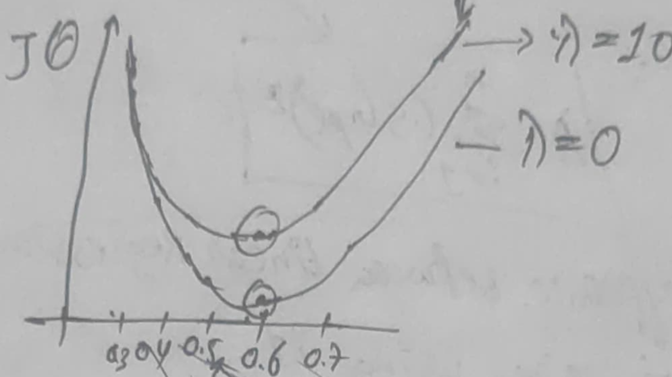
Question:- Difference between Linear Regression vs Ridge Regression!
why we are using it?
we are using it to reduce the overfitting.
How we are using it to reduce overfitting?
Because, when we add these ~~Component~~ a Component $\boxed{\lambda \sum_{i=1}^{n} (slope)^2}$ for any Slope.
~~then~~ If my (λ) Lambda value is zero. Then I will get an Gradient desent which look like this.



with the Lambda (λ) = 0.

Now,

Same 'Gradient desent how we draw in 'Linear Regession',



Lets Say,

my data points are ~~over~~ here.

Now,

As soon as we increase the (λ) Lambda we probably we try to plot θ & J θ. If we are assigning Lambda (λ) with Some positive number like $\boxed{\lambda = 10}$:

Now I will getting equation like this.

This will be $\boxed{\lambda = 10}$

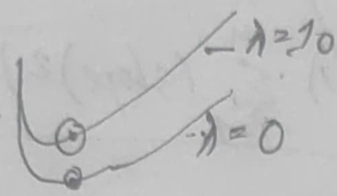
why we are discussing this directly.

Because when we set a (λ) Lambda value & if we try to slop it, we will be getting this kind of graph.

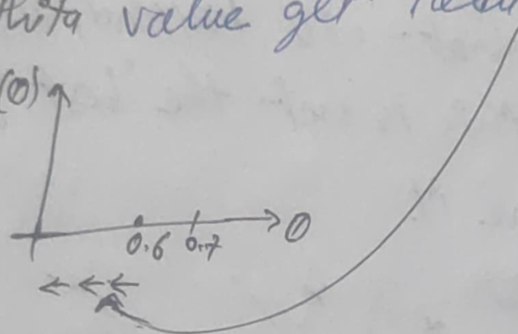
But hypothetically, it proved already.

* what are the two changes has happend: ?

(1) 1st of all my 'global minima' shifted up.



(2) my (θ) thita value get reduced. based on



my lambda (λ) value.

As the lambda (λ) value get increases my ~~the~~ (θ) thita value get decreases.

$\boxed{\uparrow(\lambda)}$ $\boxed{\theta\downarrow}$

* That is why lambda (λ) is inversely proportional to slope is (θ) thita.

In this particular Case what happend is that, If I assign some lambda (λ) value this Cost function (value) we are having some value.

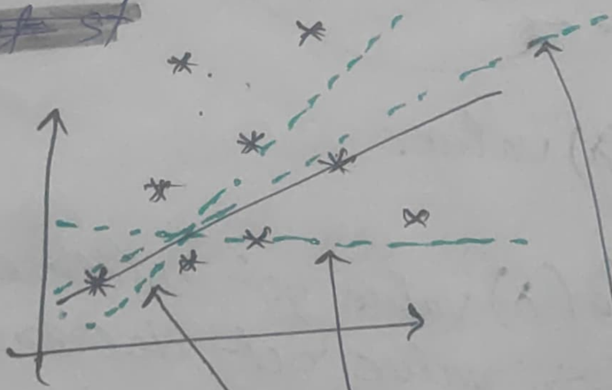
$$\text{value} + \boxed{\lambda \sum_{i=1}^{n} (\text{slope})^2}$$

But, we, will be increase some value this is

Coming from here → $\boxed{\lambda \cdot \sum_{i=1}^{n} (\text{slope})^2}$

This senario what will happend, my algorithms is says that, this is not the 'best fit line' we have to move it.

~~Next st~~



Now, next-step how will it move it, it will be moving something like this. or it may be moving something like this. or we may try to move it in a such way that, we try to create a general ratio.

That, is the region, we discussed earlier.

~~too take~~

Just take 2 data points & create a

best fit line.

In this case my · Cost function = 0

it will be zero (0).

But, if I add Ridge regression

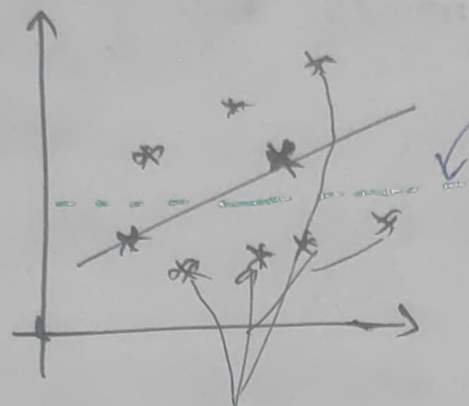
I am adding 2 parameter.

$$\text{Cost function} = 0 + \underline{\lambda(\text{slope})^2}$$

This become positive value $= (+ve)$ value ↓↓

Then my algorithms will say this is my best fit line.

or it will be keep changing like this



Since,
this is a positive (+ve) value ↓↓
it will understand that, we have to reduce
the cost ~~fun~~ function.

Now,
Instead of 'over fitting', we may get
line like this, generally it will try to cover
all the data point in a better way.
It is an assumption.

Question :-
Let's say it does not perform
well for the 'Test data' even after adding
this ?

~~Then wh~~

Then what will do!

we will keep changing the lambda (λ) value

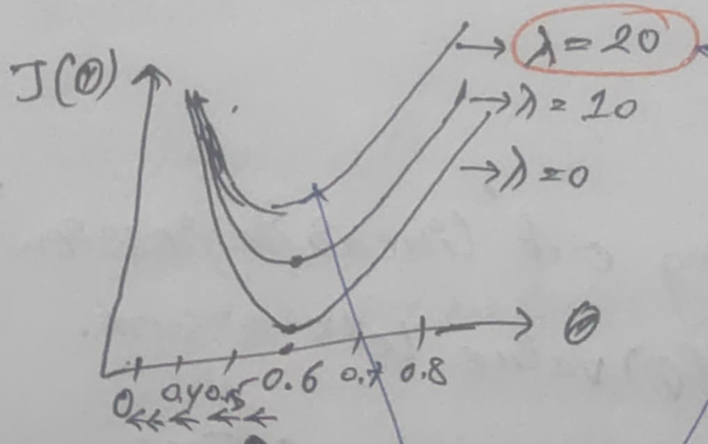
$$\text{Cost function} = 0 + \lambda(\text{slope})^2$$

& try to find out the best lambda parameter.

* If the ~~Test~~ test data continue changing, then again retraining approach is there.

* As the lambda (λ) value increases.

Let's say, lambda (λ) = 20



Now, we will be getting the line which ~~loo~~ looks like this

Here, we can see 'Global minima' has shifted. & our (θ) slope or (θ) thita value is decreasing & it going towards (0) zero.

* Question :

In Ridge Regression the thita (θ) value never become zero (0). why?

Let's say,
this is our equation.

$$\boxed{h_\theta(x) = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_3}$$

$$= \theta_0 + 0.95 x_1 + 0.82 x_2 + 1.5 x_3$$

$$\downarrow \quad \downarrow \quad \downarrow$$

$$= \theta_0 + 0.85 x_1 + 0.70 x_2 + 0.98 x_3$$

Let's say,
After finding out 'Linear Regression'
I got some thita (θ) value let's assum.

$$= \theta_0 + 0.95 x_1 + 0.82 x_2 + 1.5 x_3$$

After applying 'Ridge Regression' this thita (θ) value will decrease. this value.
It become may be like this.

$$= \theta_0 + 0.85 x_1 + 0.70 x_2 + 0.98 x_3$$

But, this will never become zero (0).

~~Question:~~

Question is that what if this coefficient become zero (0)?

$$= \theta_0 + 0.85x_1 + 0.70x_2 + \boxed{0.98x_3}$$

Let's say this coefficient become zero, what will happend.
The feature will be altogether ~~deleted~~ deleted.

$\boxed{\cancel{0.98x_3}}$.

Because there is no work of this feature.
With respect to this, since we are doing $(slope)^2$.
It will not going to become zero (0), this may reduce to at point $\boxed{0.22}$ something like this but it will be never zero.
This is the thing with respect to Ridge Regression.

* Now, we understand how ~~i~~ it reducing over fitting.

This is bassically called as 'hell to regularization'.

* Negative (-ve) value will not be there because we are ~~doing~~ squaring it. So.

(2) LASSO Regression

* Lasso Regression is basically called as [L1 Norm] & we also say it as ~~L1 Regula~~.
[L1 Regularization]

This is basically to reduce the features.
In short 'Lasso Regression' is used for 'feature selection'.

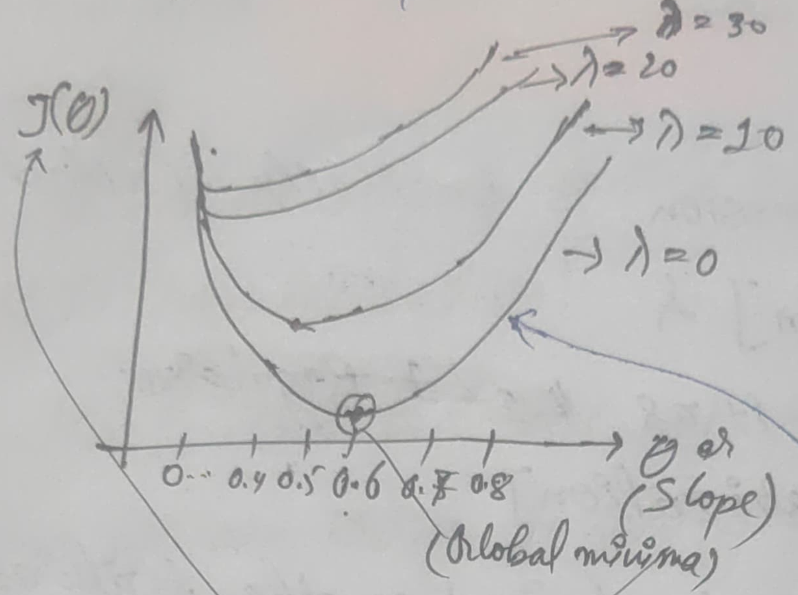
~~Cost function~~:

$$\boxed{\text{Cost function: } \frac{1}{m}\sum_{i=1}^{m}\left(h_\theta(x^{(i)})-y^{(i)}\right)^2+\lambda\sum_{i=1}^{m}|slope|}$$

2nd part here this will change that is the difference 'Ridge' & 'LASSO.'
In 'Ridge' it was $(slope)^2$ but here 'LASSO' it is going to use $|slope|$ magnitude of slope.

Question: what do we think, is the difference here & how it is reducing the features.?

There is zero is possibility to Come.



Now, if I probably find out the relationship between this ~~slope~~ ~~|slope| f(λ)~~

$\boxed{\text{Slope}} \cdot f(\lambda) \cdot (J\theta)$

using this entire Cost function; we will be seeing that,

we will initially seeing that when my λ Lambda(λ) value = 0

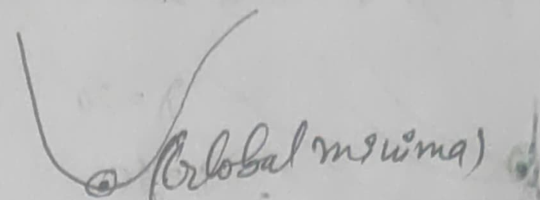
It look like this linear regression' where, I have a 'Global minima'.

This is my data points:

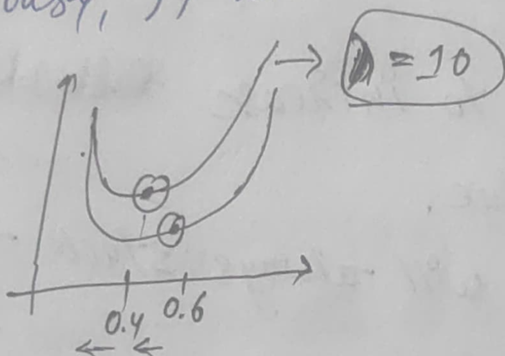
-0.2 -0.1 0 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7 ......

This is ~~wel~~ with respect to ~~lambda = 0~~

$\boxed{\text{lambda } (\lambda) = 0.}$

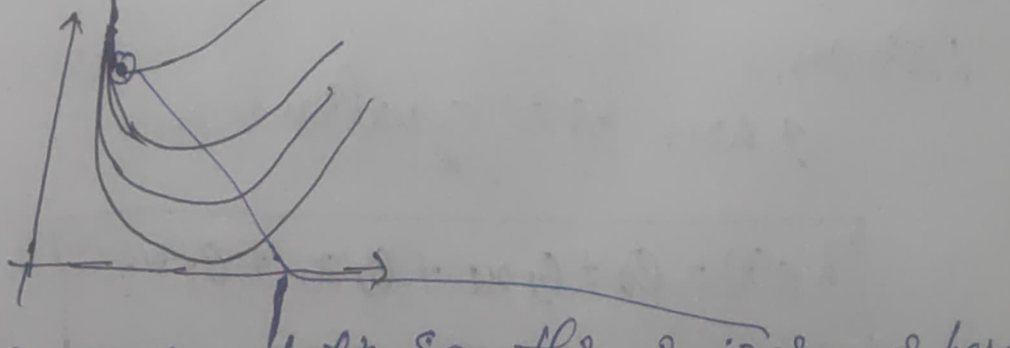


* If we increase the $\boxed{\text{lambda } (\lambda) = 10}$
obviously, it will create this kind of
curve

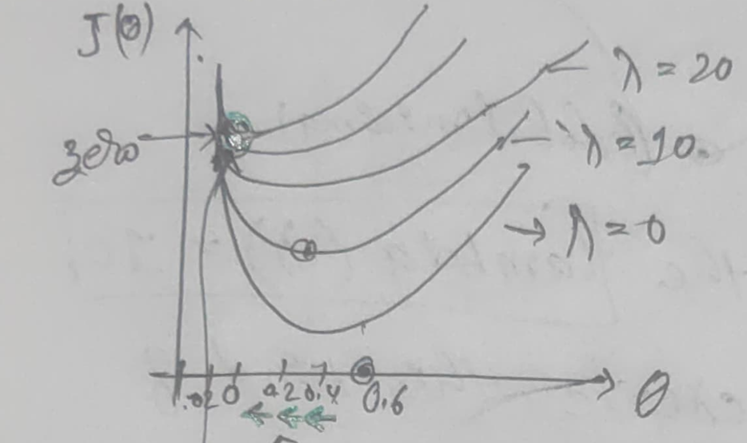


Here, our 'Global minima' shifting &
lambda (λ) is getting.

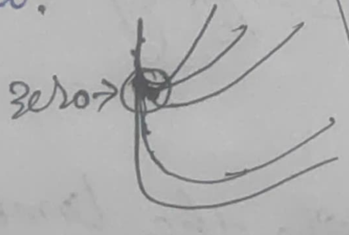
* But, if we keep increasing the lambda (λ) &
like $\lambda = 20$, $\lambda = 30$; $\lambda = 40$ value.
later we will see that, there will be a
final line which will get created.



& after zero (0). Let's say this is zero, over here then, my line will be created like this.



any time, if we try to increase ~~the line~~ the lambda (λ) value. our line will always stuck at zero.



In short our coefficient become zero.

Not all coefficient some of the coefficient.

why?

Let's say,
I have this equation

$$\boxed{h_\theta(x) = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_3}$$

* what does this equation mention ?
how many . independent feature we have.
Total $\boxed{3}$ right.

$$h_\theta(x) = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_3$$
$$= \theta_0 + 0.54 x_1 + 0.23 x_2 + 0.10 x_3$$

Let's Say,
my $\theta_0 + 0.54 x_1 + 0.23 x_2 + 0.10 x_3$
what does this .. indicate. Coefficient
with one unit movement in 'y-axis'; there is 0.54 unit movement in the ~~x-axis~~ x_1-axis.

* $\boxed{\begin{array}{l} y \uparrow 1 \\ x_1 \ \cdot 0.54 \end{array}}$

It means, if $\boxed{y}$ is increasing by $\boxed{1}$
then $\boxed{x_1}$ is increasing by $\boxed{0.54}$.

* similarly, over here $\boxed{\cdot 0.23 x_2}$

$y \uparrow 1$
x_2; 0.54. If $\boxed{y}$ is increasing by $\boxed{1}$
then $\boxed{x_2}$ is increasing by $\boxed{0.23 x_2}$.

* which is the least Corelated feature over here out ~~of~~ all this.

$$\boxed{\theta_0 + 0.54x_1 + 0.23x_2 + 0.10x_3}$$

$\boxed{x_3 \ldots 0.10}$

x_3 is the least Corealated feature over here.

$\boxed{0.10x_3}$ $(\lambda)\uparrow$

Now, with the hyper parameter tunning increasing the Lambda (λ), don't we feel it will be Come zero (0) after Some time.

If this is become zero, does this feature get deleted. or removed

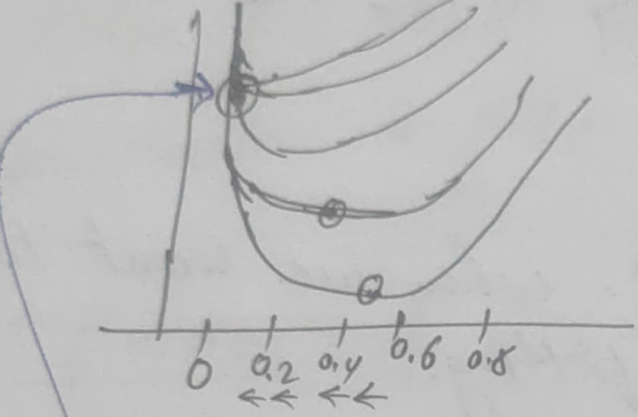
yes, automatically by the equation, without ~~doing~~ doing any thing. just playing with ~~the~~ Lambda (λ) hyper parameter.

By this equation will understand that,

$$\boxed{\theta_0 + 0.54x_1 + 0.23x_2}$$

equation ① (x₁) & ⑪ x_2 is the most important feature.

* If we go & check in our graph.



In this graph, 'Global minima' will be switching & thita(θ) also decreasing.
But this is never become zero.(0).

But here, if we keep playing with Lambda(λ) value, if we keep increasing it.
The maximum point (θ) thita will go to this zero (0)

* So, in short whichever feature are not that highly corelated or not that important automatically that will become zero.(0).
If the coefficient become zero (0) this feature $\boxed{0.10\,x_3}$ will automatically get delect or removed.

* If our data has a outliers, what we are going to use?

we will going to use the 'LASSO' Regression.

* Ridge Regression, when we want to reduce the outver fitting.

(3) ElasticNet

52:05

ElasticNet is nothing but combination of Ridge & LASSO regression.

Combination of [L1 & L_2 Norm]

$$\text{Cost function} = \frac{1}{m}\sum_{i=1}^{m}\left(h_\theta(x)^{(i)} - y^{(i)}\right)^2 + \lambda_1 \sum_{i=1}^{m}(\text{slope})^2 + \lambda_2 \sum_{i=1}^{m}|\text{slope}|$$

* 1st parameter is $\lambda_1 \sum_{i=1}^{m}(\text{slope})^2 \rightarrow$ (Ridge)
* 2nd parameter is $\lambda_2 \sum_{i=1}^{m}|\text{slope}| \rightarrow$ (LASSO)

* Note :-
ElasticNet is basically combination of both Ridge & LASSO Regression

End of the day which one is better to use we think, why do we care about outliers & why do we care about overfitting.

Let's use this together.

Cost function $= \frac{1}{m} \sum_{i=1}^{m} \left(h_\theta(x^{(i)}) - y^{(i)}\right)^2$

$\Downarrow$

MAE

$\Downarrow$

RMSE

$\Downarrow$

Huber loss

this can change to
MAE, RMSE, Huber loss.
whatever we want this is cost function
it can change.

So, the ElasticNet is the combination of both the Ridge & LASSO regression.

Interview Question :-

may asked you what is the combination of 'cost function' along with the hyperparameter tunning that we are going to provide.

---

practical start. → 55:36